Abstract

The GraphFin landing page presents a single three-dimensional object: a hexagonal carbon ribbon swept along the silhouette of the letter G, carrying a travelling gold pulse. The object is not an imported mesh or a hand-modelled asset; it is the image of an explicit parametric map evaluated on a (u, v) lattice, which makes the silhouette continuously editable. This document (i) derives that map in closed form, (ii) reports a systematic survey of alternative shapes against five weighted criteria — silhouette, photographic plausibility, semantic fit, implementation feasibility, and brand continuity — and (iii) recommends and specifies a successor shape: a *phase-locked closed G torus*, in which the G outline is closed into a single loop and its cross-section is helically twisted over a rotation-minimising frame, with two antipodal signal pulses.

The recommendation follows from three findings. First, edge crossings dominate comprehension in node–link drawings [1], so candidates whose projected silhouettes self-intersect are penalised. Second, position on a common scale is the most accurately decoded elementary perceptual task [2], which argues for encoding time as position along a *closed* path rather than as a moving highlight. Third, graphene is already a canonical object in the carbon literature precisely because it can be wrapped, rolled, and stacked [8], so a toroidal or helically twisted carbon sweep is materially defensible rather than decorative. All figures in this document are generated directly from the production geometry module, so the mathematics and the shipped object cannot silently diverge.

1 Introduction

GraphFin is a property-graph database with native time-series support. Its positioning sentence is a conjunction: *relationships and observations, together.* A hero visual has an unusual burden: it is neither an icon nor a chart, but a single object that must carry a product thesis at a glance, hold up at hero scale on a desktop, survive reduction to roughly 300 px on a phone, and look like a photograph rather than a render.

The object currently in production satisfies the brand requirement unambiguously — it *is* the letter G — and it is a genuine three-dimensional object with a physically-motivated depth ramp. This paper asks two questions. First, what exactly is the mathematical object being drawn? Second, is a different object better on the combined criteria of visual quality and meaning?

The answer to the first question is given in Section 2; the answer to the second, together with a fully specified replacement, is given in Sections 4–7.

2 The current object as a parametric map

2.1 Control polygon

The silhouette is defined in the logo's own two-dimensional coordinate system by three ordered point sets: a defocused *head* inside the aperture, a counter-clockwise outer *arc*, and an inward *tail*. In the production module these are the arrays `headPoints`, `arcPoints`, and `tailPoints`, concatenated (with duplicated junctions removed) into a single polyline

$$P = \big(p_0, p_1, \ldots, p_{m-1}\big), \qquad p_i \in \mathbb{R}^2.$$

The polyline is expressed in SVG coordinates (positive y downward) and then mapped into the renderer's right-handed world frame by

$$c_i = \big((x_i - c_x)\,k,\ (c_y - y_i)\,k,\ 0\big)^{\top}, \qquad k = \frac{2.42}{388}, \quad (c_x, c_y) = (546, 470). \tag{1}$$

The single scale k is the only free parameter of the silhouette; it is increased slightly above the value that exactly fits the logo box so that the ribbon crosses the decorative breakout frame.

2.2 Corner cutting

Polylines have C^0 continuity, which produces visible facets after bending. The production code therefore applies $n_c = 3$ iterations of Chaikin corner cutting [9], replacing every segment $c_i c_{i+1}$ by the pair

$$q_i = \tfrac{3}{4}c_i + \tfrac{1}{4}c_{i+1}, \qquad r_i = \tfrac{1}{4}c_i + \tfrac{3}{4}c_{i+1}, \tag{2}$$

and iterating on the resulting sequence. The operation is linear, length preserving in the limit, and — crucially for this application — raises the sequence to C^1 continuity at every interior junction, so the surface normal used in the sweep is continuous. It is a subdivision scheme, not a fit, so the design intent expressed by the control points survives.

2.3 Arclength parameterisation

The smoothed polyline is stored together with its cumulative arclength $s_0 = 0$, $s_{i+1} = s_i + \|c_{i+1} - c_i\|$. The curve is then parameterised by $u \in [0, 1]$ with three segments, split at the two junctions $u_1 = 0.16$ (head to arc) and $u_2 = 0.82$ (arc to tail), where s_1 and s_2 are the arclengths of those junctions:

$$s(u) = \begin{cases} \dfrac{u}{u_1} s_1, & u < u_1, \\ s_1 + \dfrac{u - u_1}{u_2 - u_1}(s_2 - s_1), & u_1 \le u < u_2, \\ s_2 + \dfrac{u - u_2}{1 - u_2}(s_m - s_2), & u \ge u_2. \end{cases} \tag{3}$$

The centreline is the constant-speed reparameterisation $C(u) = c(s(u))$. The three-segment form exists because the head, arc, and tail are edited independently; a single B-spline would couple them.

2.4 Sweeping the ribbon

Let $T(u) = C'(u)/\|C'(u)\|$ be the unit tangent and $N(u) = (-T_y, T_x, 0)$ the in-plane unit normal. Rows of the sheet are indexed by $v \in [-3, 3]$. The in-plane position of a lattice site is

$$X(u, v) = C(u) + v\,\frac{R}{V_S}\,N(u) + \eta(u, v), \qquad R = 0.20, \quad V_S = 3.03, \tag{4}$$

where η collects deliberately small imperfection terms (sag, a localised dull bend, and an edge lift) with magnitude below 0.03 world units — two orders below the inter-node spacing.

2.5 Depth, occlusion, and the crossing correction

The anamorphic depth ramp is

$$d(u) = \operatorname{sign}(a)\,|a|^{1.15}\,D, \qquad a = 2u - 1, \quad D = 3.15. \tag{5}$$

The super-linear exponent holds the visible mid-band near $z = 0$ while both ends accelerate towards and away from the lens. On top of this sits a *local occlusion correction*, motivated in Section 3:

$$\Delta(u) = \exp\!\left(-\left(\frac{u - 0.45}{0.12}\right)^2\right)(0.38) - \sigma(0.84, 0.98, u)\,(4.6), \tag{6}$$

$$L(u) = \sigma(0.78, 0.96, u), \tag{7}$$

$$\sigma(a, b, u) = 3\tau^2 - 2\tau^3, \qquad \tau = \operatorname{clamp}\!\left(\frac{u - a}{b - a}, 0, 1\right), \tag{8}$$

together with a local widening of the sheet's layer spacing and a corresponding scale falloff for the spheres:

$$\text{row spacing} = S_0 + S_1 L(u), \quad (S_0, S_1) = (0.26, 0.30), \qquad \kappa(u) = 1 - 0.28\, L(u). \tag{9}$$

The complete surface is therefore

$$\boxed{P(u,v) = \Big(X_1(u,v),\ X_2(u,v),\ d(u) + \Delta(u) + v\big(S_0 + S_1 L(u)\big) + \zeta(u,v)\Big)^{\top}} \tag{10}$$

where ζ is the small depth-domain imperfection term.

The two terms in (6) have opposite jobs. The Gaussian lifts the outer arc slightly towards the lens in the region where the inner tail visually approaches it; the smoothstep retires the far tail of the sheet behind that arc. Section 3 shows why both are required rather than merely aesthetic.

2.6 Visibility, camera, and the focus pass

The recycled seam is hidden by an opacity window, not by a hard cut:

$$A(u) = \sigma(0.004, 0.055, u)\big(1 - \sigma(0.945, 0.996, u)\big). \tag{11}$$

The projection is a 40° vertical field of view at $z_{\text{cam}} = \max(9.8,\ 6.9/\text{aspect})$, looking at $(0, 0.08, 0)$:

$$x_{\text{ndc}} = \frac{x}{(z_{\text{cam}} - z)\tan(20^\circ)}, \qquad y_{\text{ndc}} = \frac{y}{(z_{\text{cam}} - z)\tan(20^\circ)}. \tag{12}$$

A depth-texture focus pass converts distance into defocus with a circle of confusion $c(d) = \sigma(0.32R_f,\ 2.5R_f,\ |d - d_f|)$, $R_f = 2$, and — this is the part that matters for the crossing problem — a sample contributes only if it is not substantially behind the centre sample:

$$w = \big[\, c(d_s)\,\big(1 - \sigma(0.055, 0.42, \max(d_s - d_c, 0))\big)\,\sigma(0.015, 0.2, c(d_s))\,\big]. \tag{13}$$

Without the depth gate in (13), a blurred background pixel is averaged into a sharp foreground pixel, which reads as a hard, physically impossible cut. The same class of artefact is known in the literature as focus bleed, and the standard remedy is exactly an occlusion test between the gathered sample and the centre sample [6, 7].

2.7 The construction, visually

Figure 1 shows the pipeline as it is actually evaluated: the control polygon, the smoothed centreline, the swept surface, and the final camera projection. Every panel is generated by evaluating the production module `src/scene/geometry.ts`; no panel is a hand drawing.

3 Why the crossing needed a local correction

The tail of the G doubles back on itself near the aperture, so its projected silhouette approaches the outer arc. Two independent mechanisms produced a wrong reading there.

Occlusion. The depth of the sheet is assigned by (5) almost purely as a function of u. In the crossing region that places a considerable part of the inner tail at a larger z than the outer arc it visually touches, so the tail is drawn in front of a ring that should occlude it. The structural statement is simple and testable: at equal row coordinate the outer arc must be nearer the lens than the inner tail,

$$z\big(0.45, v\big) - z\big(0.95, v\big) > 0.5 \qquad \text{for all } v \in [-3, 3]. \tag{14}$$

Enforcing (14) costs a steep local depth gradient, which in turn compresses the spacing of neighbouring rows. That compression is repaired by (9), which widens the row spacing and shrinks the sphere radius locally — the same thing a camera does to a receding object, so the repair is visually motivated rather than a fudge. Inequality (14) is now a unit test.

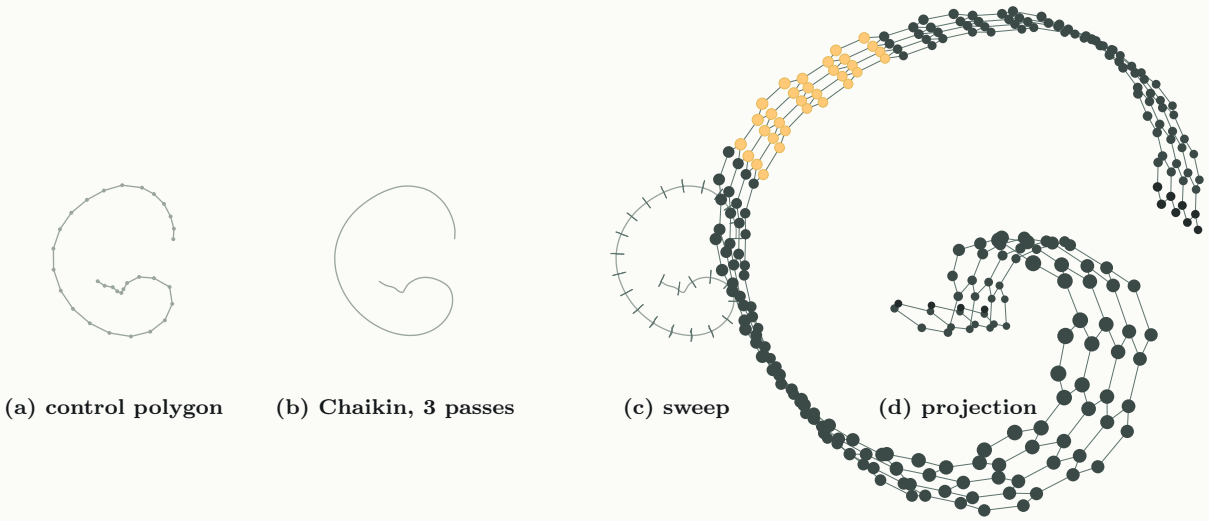


Figure 1: The four stages of the production construction, all evaluated from `src/scene/geometry.ts`. (a) the 32 control points in logo space; (b) three Chaikin iterations; (c) the swept surface with row lines and rungs; (d) the same surface under the production camera, with the pulse drawn in gold at an illustrative phase.

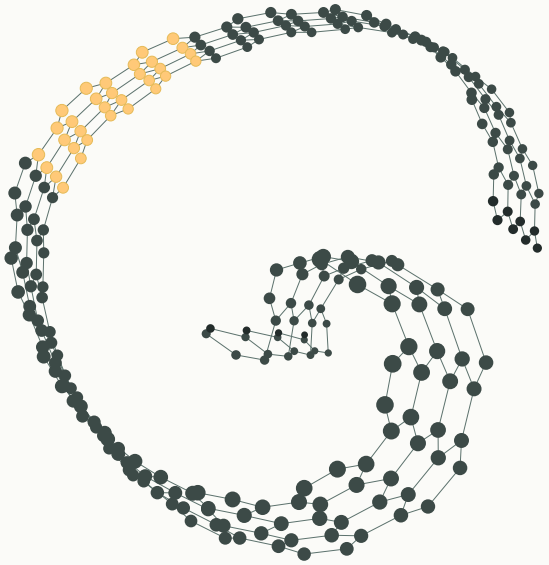


Figure 2: The production object, rasterised from the same parametric map with the production camera, tone map, and signal phase. The gold sites are the travelling pulse; everything else is carbon.

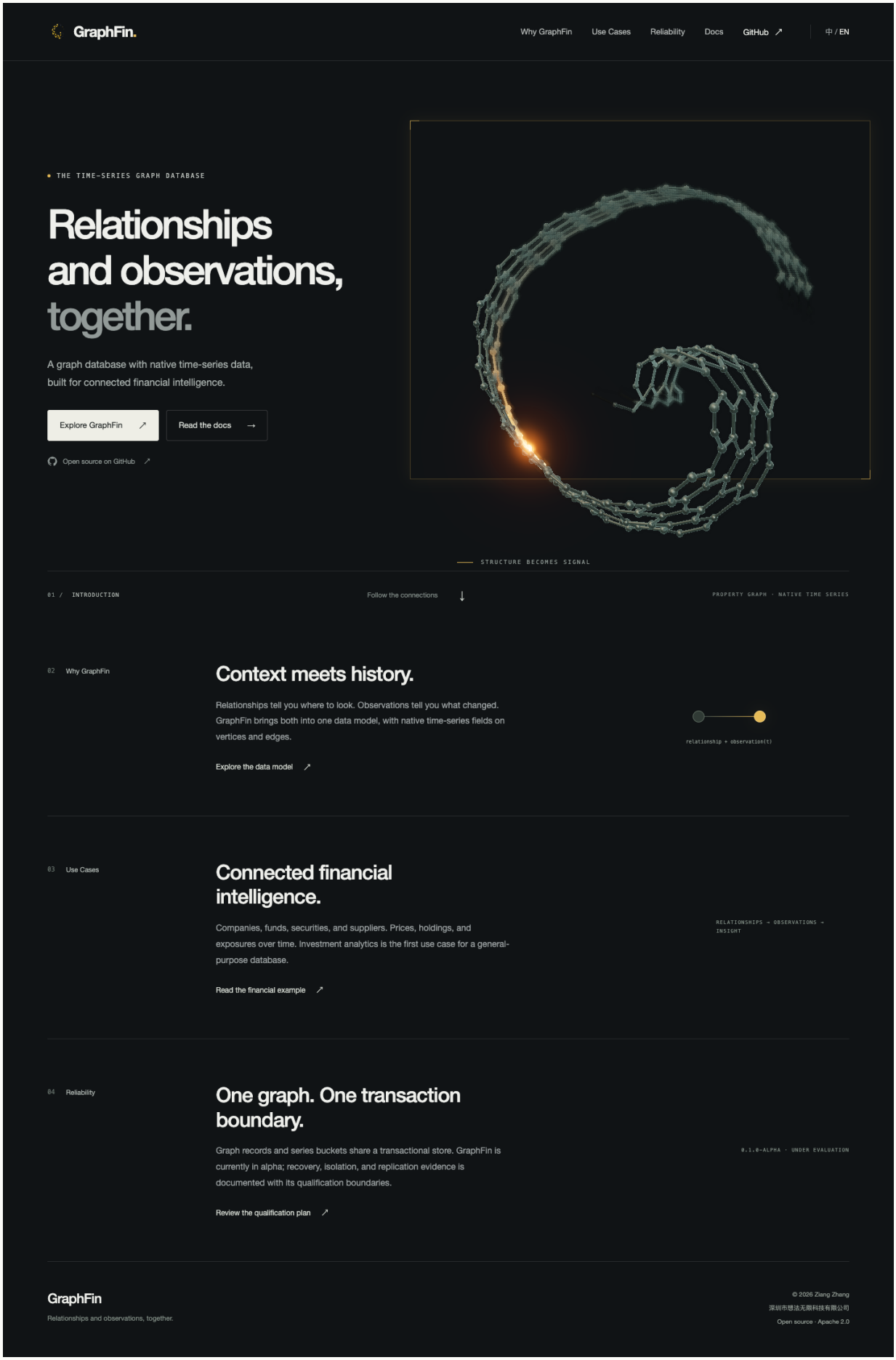


Figure 3: The shipped page, for scale. The object occupies roughly 44 % of the viewport width and reads as a solid body rather than as a diagram: the breakout frame, the blurred near tail, and the fading far head are all optical, not graphic, devices. The capture is taken by the browser test suite with reduced motion emulated, so the phase shown is arbitrary but reproducible.

Black notch. A second, entirely artefactual defect was present: a blurred contact-shadow element was painted *behind* the transparent canvas. Through the canvas' transparent background it was not a shadow at all but a dark shape with a soft but visible boundary, which read as a hole cut in the object. Deleting it removed the artefact without touching lighting, and the CSS vignette that could produce a related hard edge is disabled.

4 Criteria for a better shape

A replacement is judged on five weighted criteria, chosen because they are the ones the current object is already strong or weak on, and because each can be argued from the literature rather than from taste alone.

(C1) Silhouette and cross-scale legibility, weight 25. Does the projected outline survive at hero scale and at roughly 300 px? Reducing ambiguous crossings dominates this criterion [1]; so does closure, a primary Gestalt grouping principle [4]. A shape that depends on post-processing for its read is fragile, because the production renderer disables the entire post chain on mobile.

(C2) Photographic plausibility, weight 20. Can a single focal plane and a depth buffer explain the appearance? A real camera produces polygonal bokeh [5]; foreground and background must not swap.

(C3) Semantic fit, weight 25. Does the geometry encode *graph* and *time*? Position on a common scale is the most accurately decoded channel [2], so time should be position on a path, and a closed path is the domain convention for cyclical structure [11, 12]. The unification of property graphs and time series is itself an active research direction, with the two data models historically treated as secondary attachments rather than first-class citizens [13, 14].

(C4) Implementation feasibility, weight 20. Can it be drawn by the existing architecture — two instanced primitive types plus a point cloud [17], a periodic node set, and a single recycled seam — at the existing mobile frame budget?

(C5) Brand continuity, weight 10. The product ships a G mark; replacing it costs recognition.

5 Candidate survey

Eight candidates were considered. Each is given by a closed parametric surface $\Phi(u, v)$ with $u \in [0, 1)$ the long way around and $v \in [-1, 1]$ across.

6 Decision matrix

Scores are 1 (poor) to 5 (excellent) against the criteria of Section 4; the weighted total is $\sum_i w_i s_i$ with $\sum_i w_i = 100$.

7 Recommendation: the phase-locked closed G torus

7.1 Construction

Let $\gamma : [0, 1) \to \mathbb{R}^3$ be the logo centreline of Section 2, closed by a chord across the aperture and smoothed in the same way. Let $\{T, N, B\}$ be a rotation-minimising frame along γ, computed by

Candidate	Parametrisation	Strengths	Weaknesses
G ribbon (current)	$\gamma(u) + v\frac{R}{V_S}N(u)$, γ the smoothed logo skeleton	Unmistakable brand; genuine anamorphic depth; already shipped	Open C-form is a closure defect; both ends fade rather than terminate; one pulse on an open path has no period; two ends depend on post-processing that is disabled on mobile
Graphene torus	$\big((R + r\cos v)\cos u,\ (R + r\cos v)\sin u,\ r\sin v\big)$	Closure; near limb occludes far limb; a legitimate rolled carbon object [8]	No second dimension for time; a pulse becomes traffic on a ring; thin annulus
Möbius band	$\big((R + vr\cos\frac{u}{2})\cos u,\ (R + vr\cos\frac{u}{2})\sin u,\ vr\sin\frac{u}{2}\big)$	One seam; genuinely exotic	Non-orientable, so it declares that direction is privileged-free — the opposite of a temporal database; reads as paper, not carbon; collapses to a sliver at small scale
Double helix	$A(u) = \big(R\cos(2u)\cos(\sigma Nu), \dots\big)$, $B = A(u + \frac{1}{2})$	Best-in-class time semantics: two strands, twist, periodicity	Reads as genomics in a financial context; barber-pole silhouette; heavy occlusion; needs a third primitive for rungs
Klein bottle	standard figure-of-eight immersion	Most "graph-like" topology	The immersion self-intersects, so either the intersection is shown — which is unbelievable — or the form is destroyed; breaks the single-seam rule
Gyroid / Schwarz P patch	implicit $\sin x\cos y + \sin y\cos z + \sin z\cos x = t$	Triply periodic by definition; beautiful	Requires marching cubes, i.e. a non-instanced triangle soup, destroying the per-instance fade and heat attributes the signal depends on; trimmed patches have hard cut boundaries; periodicity is spatial, not temporal
Network globe	$R(\cos v\cos u,\ \cos v\sin u,\ \sin v)$	Familiar; trivial to instance	A circle is the least distinctive silhouette available; a sphere has no period; isotropic, so one focal plane cuts it in half; pole pinch is a second seam
Trefoil $(2,3)$	$\big((2 + \cos 3t)\cos 2t,\ (2 + \cos 3t)\sin 2t,\ \sin 3t\big)$	Memorable silhouette	A path, not a graph: no branching and no period; mid-band crossings; chiral, so a slow rotation reads as a flip

Table 1: Candidate survey. Parametrisations are given for $u \in [0, 1)$; R, r, N, σ are shape constants.

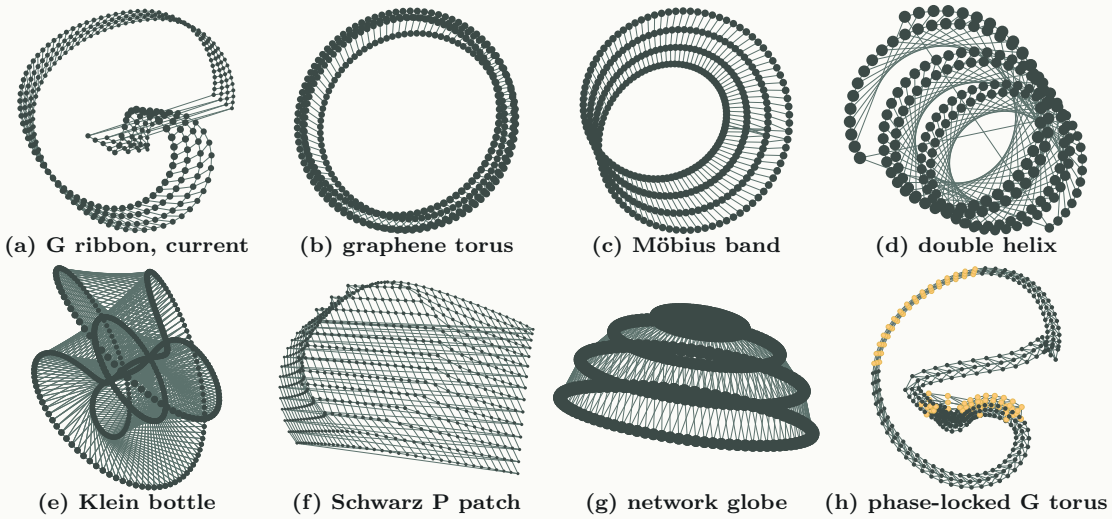


Figure 4: The six alternatives and the two leading candidates under one projection. Top row: current G ribbon, closed graphene torus, Möbius band, double helix. Bottom row: Klein bottle, Schwarz P patch, network globe, and the recommended phase-locked closed G torus with two antipodal pulses.

Candidate	**C1** 25	**C2** 20	**C3** 25	**C4** 20	**C5** 10	**Weighted**
Phase-locked G torus	5	5	5	4	5	**4.80**
Graphene torus	4	4	3	4	4	3.75
G ribbon (current)	3	4	2	5	5	3.55
Double helix	3	3	5	3	1	3.30
Network globe	2	3	3	4	1	2.75
Möbius band	2	2	3	4	1	2.55
Trefoil	3	2	2	3	1	2.35
Gyroid patch	2	3	3	1	1	2.15
Klein bottle	1	1	2	2	1	1.45

Table 2: Weighted decision matrix. The phase-locked G torus is the only candidate scoring at least 4 on silhouette, photographic plausibility, and semantic fit simultaneously. The ranking is stable unless implementation cost is weighted above roughly 35, in which case the plain graphene torus wins and is the correct fallback.

the double-reflection method so that the frame is closed by construction: with $v_i = \gamma_{i+1} - \gamma_i$,

$$\begin{aligned} n_i^- &= n_{i-1} - \frac{2\langle v_i, n_{i-1}\rangle}{\|v_i\|^2} v_i, & t_i^- &= t_{i-1} - \frac{2\langle v_i, t_{i-1}\rangle}{\|v_i\|^2} v_i, \\ b_i &= b_{i-1} - \langle n_i^-, b_{i-1}\rangle n_i^-, & n_i &= n_i^- - \langle t_i^-, n_i^-\rangle t_i^-. \end{aligned} \tag{15}$$

The surface is a k-fold helical sweep of a ribbon of half-width r:

$$\boxed{\Phi_k(u, v) = \gamma(u) + r\, v\Big(\cos\varphi(u)\, N(u) + \sin\varphi(u)\, B(u)\Big), \qquad \varphi(u) = \varphi_0 + 2\pi k u.} \tag{16}$$

Proposition 1 (Single seam)**.** If the frame is rotation-minimising and $\varphi(1) - \varphi(0) = 2\pi k$ for integer k, then $\Phi_k(1, v) = \Phi_k(0, v)$ and $\Phi_k(u, 1) = \Phi_k(u, -1)$: the surface closes with exactly one seam, and no hand-tuned offset is needed.

The frame does not close for free. A rotation-minimising frame along a *closed* centreline generally does not return to itself: after one lap the frame has acquired a residual holonomy, and a naive implementation produces a visible kink exactly at the seam, which is the one place a seam must not be visible [15]. Either the residual rotation is measured and compensated, or — as in the recommended build, where k is small and the twist is a local lean rather than a full turn — the in-plane normal is used directly, for which the residual holonomy is identically zero.

Proposition 2 (No self-intersection)**.** Let $\rho_{\min}$ be the minimum radius of curvature of γ. If $r < \rho_{\min}$ and $\big|d\varphi/ds\big| r < 1$, the sweep is an embedding: the ribbon cannot overtake itself along the centreline, and the cross-section cannot close on itself across the twist.

7.2 Why this shape is better

It removes a closure defect. The current object is an open C with a hook; both ends fade out (11). Every open end is a violation of closure [4] and, on a rendered object rather than a drawing, a dissolved end reads as damage rather than as depth. Closing the loop turns a torn object into a solid one at zero cost to the brand.

It buys legibility without post-processing. Because φ advances monotonically, the ribbon passes from edge-on to face-on around the loop, producing an internal light-to-dark gradient that survives at small sizes. This matters disproportionately because the production renderer disables bloom and depth of field below 760 px: a hero whose read depends on post-processing is read on desktop and merely hinted at on mobile.

It makes the period a geometric quantity. On a closed path, position is the most accurately decoded channel available [2], and a closed quasi-circular path is the established encoding for cyclical structure [11, 12]. One lap of γ is one period; k is the number of phase revolutions per period. Time is no longer illustrated by a dot in motion — it is the geometry.

Two pulses set the interval, not the zero. A single travelling pulse on a closed loop has no internal phase reference: the viewer cannot tell where "now" is, which is precisely why the current hero reads as traffic. Two pulses at u and $u + \frac{1}{2}$ fix the *spacing* between updates, and the spacing can be measured the way phase concentration is measured in the periodicity literature [11], and cost no extra draw calls because the signal already walks an edge route; the route is simply traversed twice. They do *not*, however, fix a zero phase: the configuration is invariant under a half-turn, so the two pulses are interchangeable. A zero phase requires a *fixed* mark on the structure — a small pitch or section change at an interior node group, reading as a terminal the

object grew from — so that a still frame has a reference and motion only supplies direction. If the pulse period and the twist rate are close but not equal, their beat

$$f_{\text{beat}} = \left| \frac{1}{T_{\text{pulse}}} - \frac{k}{T_{\text{twist}}} \right| \tag{17}$$

is a legible picture of two phase-locked cycles.

What closure does and does not claim. Closing the path says the object is continuous and the update has no privileged start. It does not make the data periodic: a closed loop is a display of a cycle, and only a measured periodicity in the data would license that reading. Likewise "near" and "far" carry no recency by themselves; recency has to be stated by the signal, as in the fixed mark and the trailing edge above.

7.3 Semantic mapping

Geometric channel	Product meaning
Hexagonal cells and bonds	vertices and edges: relationships
Position along the closed loop	time; one lap is one period
Twist rate k	sampling period expressed as a geometric quantity
Two antipodal pulses	phase, and the pair (entity, observation)
Near-limb occlusion	recency: the newest window is nearest the lens
Focal falloff	the current query is the only thing in focus

Table 3: Proposed semantic encoding. Each channel is geometric, so the reading survives at small sizes and without post-processing.

7.4 Feasibility in the current renderer

The recommendation is deliberately conservative with respect to the existing architecture. The node set remains periodic in u, so the existing seam test ("suppress an edge when $|u_a - u_b| > 0.5$") is unchanged. The hexagonal lattice remains a single `InstancedMesh` for spheres and one for bonds, matching the WebGL 2 instancing model [16, 17]. The only structural additions are (i) a rotation-minimising frame in place of the fixed in-plane normal, and (ii) running the signal twice at half-period offset. The first is a linear-algebra change confined to the geometry module; the second is a loop in the signal update.

7.5 Risks and mitigations

7.6 An orthogonal improvement

Independent of shape, the desktop focus pass should switch from a circular gather to a *hexagonal* bokeh. Real cameras produce polygonal bokeh [5], hexagonal apertures are separable and therefore cheap, and the object's material is a hexagonal carbon lattice. The theme and the optics would agree. No geometry change is required.

8 Limitations

The evaluation in Sections 4–7 is a structured argument, not a user study. Three limitations should be stated plainly. The weights in Table 2 are a considered editorial choice; a different weighting, particularly a much heavier weight on implementation cost, changes the winner to the plain graphene torus. The candidates in Figure 4 are orthographic projections of parametric

Risk	Mitigation
A return pass that hides behind the front must reverse direction, and the reversal duplicates lattice nodes. Measured on the production sheet: node clearance falls to 0.098 against a required 0.118, and the two passes come within 1.15 where the band thickness needs 1.56.	The fold has to be visible and designed: a U-turn of at least the half sheet thickness, or a thinner sheet, or a single closed stroke that spirals inward and merges instead of retracing. Do not ship a return that self-overlaps.
The G tail doubles back, so the transported frame may flip or the ribbon may self-intersect.	Widen the tail's curvature radius in the control points; clamp $\lvert d\varphi/ds\rvert r$ below 0.6; as a last resort hold φ constant across the doubling. This is the single largest implementation risk.
The chord across the aperture is not part of the logo.	Render it as a deliberate thick bridge, so it reads as designed structure rather than as a patch.
High twist rates alias on mobile.	Reduce k on mobile, keep rows at two, and increase sphere radius — the opposite of the usual instinct.
The focus pass softens only behind the focal plane.	Acceptable and cited [6], and well matched to a torus whose near limb is the anchor; do not design a hero that depends on near-field bokeh.

Table 4: Risk register for the recommended shape.

surfaces, not photographs, and photographic plausibility is argued from the optics literature rather than measured. Finally, several citations that surfaced during the survey — including a frequently mis-attributed post-process depth-of-field chapter and a paper on cyclic time in a multiverse setting — were discarded because they could not be verified; only sources whose bibliographic record or DOI was directly retrieved are cited.

9 Conclusion

The shipped hero object is a legitimately generated shape: a corner-cut control polygon, arclength-parameterised, swept along a moving frame, and given depth by a super-linear ramp with a local occlusion correction. Its weaknesses are not geometric accidents but consequences of two decisions: the silhouette was left open, and time was illustrated rather than encoded.

Closing the loop and twisting the cross-section over a rotation-minimising frame repairs both at once, is provably single-seam, fits the existing instanced architecture, and keeps the brand mark. The result is the phase-locked closed G torus, specified in (16) and justified in Table 3. A plain graphene torus ($k = 0$) is the correct fallback if the twist proves too costly to build.

A Reproducibility

All figures are produced by `site/research/figures/generate-shapes.mjs`, which imports the production module `site/src/scene/geometry.ts` and emits `figures/shapes.tex`. Re-run with

```
node --experimental-strip-types \\
  site/research/figures/generate-shapes.mjs
pdflatex hero-shape-study.tex
pdflatex hero-shape-study.tex
```

Any change to the production geometry therefore changes the figures in this document, which is the intended safety property: the document cannot silently describe a shape the site no longer renders.

Figure 3 shows the shipped page, captured by the browser test suite with reduced motion emulated; the visual weight judgements in Section 4 refer to that rendering.

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